

DAE-Net: Dual Attention Mechanism and Edge Supervision Network for Image Manipulation Detection and Localization

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Abstract—Image manipulation detection and localization (IMDL) aim to identify tampered regions from a suspicious image. Although existing works have achieved impressive results, they fail to capture sufficient tampering artifacts resulting in inaccurate localization of tampered regions. In addition, the problem of poor robustness to postprocessing methods also occurs frequently. To address these problems, this article presents a dual attention mechanism and edge supervision network (DAE-Net) to locate forged regions. Specifically, DAE-Net first extracts multiscale semantic features from RGB images and captures global and local inconsistent noise features from noise maps. Subsequently, to mine the complementary relationship among different domains and obtain more discriminative feature representations, a dual attention mechanism (DAM) module is designed to integrate semantic features and noise features. Then, we exploit a mask generation module (MGM) consisting of fusion upsampling module (FUM) to generate tampered regions in a progressive decoding manner. FUM fully fuses multiscale semantic features to enhance tampering traces. Finally, an edge supervision module (ESM) is introduced to capture subtle edge information to optimize the boundary of tampered regions. The experimental results on five public datasets demonstrate that DAE-Net outperforms the state-of-the-art (SoTA) methods in terms of higher localization accuracy and stronger robustness under widely used evaluation metrics. F_1 /AUC of the proposed DAE-Net reach 0.494/0.805, 0.871/0.984, 0.972/0.973, 0.330/0.741, and 0.338/0.826 on CASIA, NIST16, Columbia, Coverage, and IMD2020, respectively.

Index Terms—Dual attention mechanism (DAM), edge supervision, image forensics, image manipulation detection and localization (IMDL), progressive decoding manner.

I. INTRODUCTION

NOWADAYS, with the emergence of various user-friendly image editing tools, people can easily modify digital images [1]. These tampered images appear realistic and have no obvious difference from the common images [2]. When tampered images appear in application scenarios that are extremely sensitive to image content, such as news media, academic research, and military communication [3], this will cause a trust crisis and threaten social stability. Therefore, it is necessary to develop trustworthy models for identifying forged images.

In general, image manipulation techniques can be divided into two categories: content-dependent process and content-independent process [4]. The former is more harmful than the latter since it alters the semantic contents of the image and is more misleading. The content-dependent process includes copy-move [5], splicing [6], and removal [7], as shown in Fig. 1. The content-independent process includes brightness change, blurring, nosing, and image compression [4]. They make global modifications and barely create any disinformation, but they may hide visual clues and discrepancies between authentic and manipulated regions.

The issue of image manipulation detection and localization (IMDL) has received widespread attention. Many researchers have proposed different methods for IMDL in the past. In the early stages, conventional IMDL methods relied on hand-crafted features, such as feature point matching [8], color filter array (CFA) [9], noise inconsistency [10], discrete wavelet transform (DWT) [11], discrete cosine transform (DCT) [12], and JPEG compression artifact [13]. These methods are dependent on prior knowledge and can deal with specific types of tampering techniques. Nevertheless, when facing with complex and unknown tampering types, the accuracy and robustness of these methods are insufficient for practical applications. Recently, deep learning has demonstrated strong performance in computer vision tasks, such as image segmentation [14], image classification [15], and object detection [16], because of its adaptive feature extraction capability. Therefore, deep learning techniques have also been applied to address the issue

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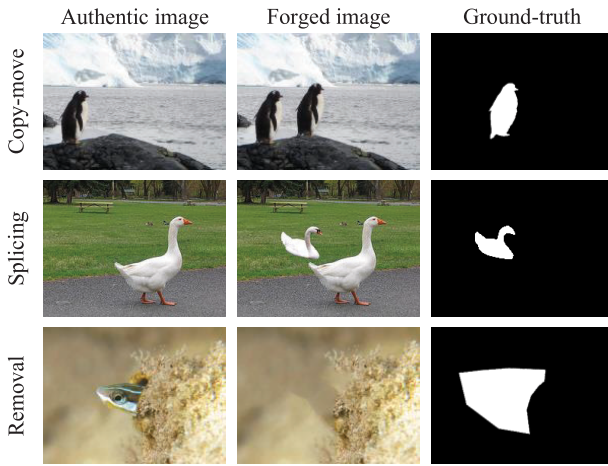


Fig. 1. Examples of content-changed-based image manipulation techniques, i.e., copy-move, splicing, and removal. The first column is the authentic image, the second column is the forged image, and the third column is the ground-truth mask.

of IMDL. A portion of methods are designed to handle a single type of manipulation, such as image copy-move forgery detection [17], image splicing detection [18], and image inpainting detection [19], which are not suitable for general IMDL. To detect multiple manipulation types, generic IMDL models have been proposed, such as ManTra-Net [20], RGB-N [21], and SPAN [22]. These methods can efficiently extract tampering artifacts and achieve manipulation localization at the pixel level.

These methods can be divided into two categories based on the network structure: single branch networks and multi-branch networks. Single branch networks [3], [23] directly process the forged images and exploit convolutional neural network (CNN) to extract the RGB features for localizing tampered regions. This could lose the tampering artifacts that cannot be captured in the RGB domain, resulting in false localization of tampered regions. Multibranch networks utilize multiple CNNs in parallel or sequential order to capture diverse features to realize accurate localization, such as noise features [21], [24], frequency-domain features [25], [26], and edge features [2]. These approaches based on noise features can enhance tampering traces and suppress image semantic content. Noise maps are extracted in three main categories: steganalysis rich model (SRM) filters [21], Bayar convolution (BayarConv) [27], and combined convolution [28]. To achieve more refined tampered boundaries, methods [2], [24] of the edge optimization are proposed.

Although the aforementioned approaches can perform general image manipulation detection and achieve tampered region localization at the pixel level. However, there are still three unsolved problems for IMDL: 1) image forensic approaches based on CNN lose the image semantic information with the deepening of network, resulting in inaccurate manipulation localization in the process of mask generation; 2) multibranch methods perform simple concatenation only in the channel dimension when different features are fused, failing to explore the complementary relationship of the features in different domains and to achieve effective feature

fusion; and 3) existing methods suffer from rough and incomplete boundaries due to ignoring or destroying subtle edge artifacts. However, the existing methods cause degradation in detection performance when detecting some forged images that have been processed by high noise or high compression. This is because forged images after processing by high noise and high compression will seriously damage the tampering artifacts of the image and the inconsistency of the boundaries between the forged and real regions, which makes it difficult for the model to extract effective feature representations. In addition, since the size of the tampered area is generally random, which requires the model to have a strong sensitivity to the scale transformation of the tampered area. Since small tampered areas produce sample imbalance and contain fewer tampering artifacts, most of the existing image manipulation detection methods mainly focus on how to use the network to better extract the tampered features, ignoring the sample imbalance caused by small tampered areas, which affects the improvement of the model's detection performance.

To improve the accuracy of manipulation localization and enhance the robustness against image postprocessing operations, we propose a dual attention mechanism and edge supervision network (DAE-Net) for IMDL, as shown in Fig. 2. DAE-Net focuses on detecting three common manipulated techniques, i.e., copy-move, splicing, and removal. Compared with the existing methods, the proposed DAE-Net exploits multiple tampering artifacts for IMDL. Specifically, we introduce an RGB branch to extract multiscale semantic features for tampered regions prediction and edge information extraction. Meanwhile, the noise branch (NB) employs the backbone of swin transformer (SWT) [29] to capture noise features that fuse global and local inconsistent information. To obtain more discriminative feature representations, a dual attention mechanism (DAM) module is designed to mine the complementary relationship among different domain features by computing the self-attention in the spatial and channel dimensions. Subsequently, the mask generation module (MGM) consisting of a fusion upsampling module (FUM) localizes tampered regions in a progressive decoding manner. The FUM exploits multiscale semantic information to improve the accuracy of localization. Finally, to optimize the boundary of tampered regions, the edge supervision module (ESM) is introduced to enhance the subtle edge features and guide the network to predict more refined tampered regions. Extensive experiments are conducted on five public datasets. The experimental results show that the proposed DAE-Net outperforms state-of-the-art (SoTA) methods.

In summary, the main contributions of this work are as follows.

- 1) We propose a DAE-Net for IMDL, i.e., DAE-Net, which extracts multiple tampering clues to more accurately localize tampered region, including semantic features, fused global and local inconsistent noise features, and edge features.
- 2) A DAM module is designed to mine the complementary relationship between semantic features and noise features. This is achieved by computing self-attention

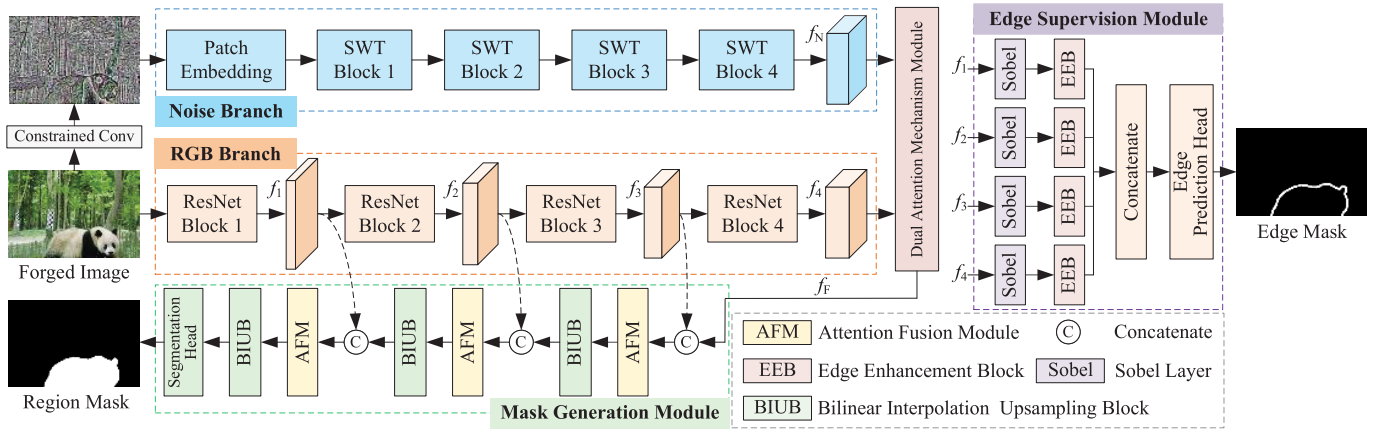


Fig. 2. Overview of the proposed DAE-Net for image manipulation localization. The input is a suspicious image ($H \times W \times 3$), and the output is a predicted mask ($H \times W \times 1$), which localizes the tampered regions. DAE-Net consists of five components: RGB branch, NB, DAM module, MGM, and ESM.

TABLE I
SUMMARY OF SOTA METHODS FOR IMDL

Category	Method	Forensic Clues		Backbone	Attention	Scales of Supervision		
		RGB	Noise			Pixel	Image	Edge
Unsupervised	CFA1 [9]	+	Local CFA	-	-	+	-	-
	NOI1 [10]	-	Noise	-	-	+	-	-
DNN-Based	ManTra-Net [20]	+	BayarConv, SRM Filter	Wider VGG	-	+	-	-
	SPAN [22]	+	BayarConv, SRM Filter	Wider VGG	Spatial Pyramid Attention	+	-	-
	DenseFCN [30]	+	-	Dense Block	-	+	-	-
	SATFL [31]	+	-	Wider VGG	Forgery Attention	+	-	-
	PSCC-Net [4]	+	-	HR-Net	Spatio-Channel Correlation	+	+	-
	TB-Net [25]	+	DCT	ResNet101	Adaptive Cross-Attention	+	-	-
	TA-Net [2]	+	-	Res2Net50	Reverse Attention Mining	+	-	+
	DMFF-Net [32]	+	BayarConv	ResNet50	Feature Fusion Module	+	-	-
	SF-Net [33]	+	DCT	HRNet	Spatial-Channel Fusion Excitation	+	-	+
	MVSS-Net [24]	+	BayarConv	ResNet50	Channel Attention, Position Attention	+	+	+
	DAE-Net (ours)	+	BayarConv	ResNet50, Swin-Transformer	Dual Attention, Attention Fusion Module	+	-	+

in both spatial and channel dimensions, thus enabling sufficient fusion of features from different domains.

- 3) In the mask generation stage, a FUM is proposed to integrate multiscale semantic features to accurately localize tampered regions in a progressive decoding manner. Meanwhile, an ESM is utilized to optimize the boundary of tampered regions.
- 4) Experimental results on five public datasets demonstrate that the proposed DAE-Net outperforms the SoTA methods in terms of tampered localization accuracy and model robustness.

The rest of this article is organized as follows. Section II presents the related work on IMDL. The proposed approach is described in detail in Section III. Section IV presents the experimental results and corresponding discussions. Finally, Section V concludes this article and discusses current limitations and future research directions.

II. RELATED WORKS

In this section, we review related work on deep neural network (DNN)-based IMDL. The methods for attention

mechanism and edge supervision are introduced, and the differences between this work and the existing work are pointed out. The related work is briefly summarized as shown in Table I.

A. Image Manipulation Detection and Localization

IMDL is one of the research topics in digital media forensics. Typical image manipulation techniques based on content changes can be divided into three categories: copy-move, splicing, and removal. Most of the early work focused on detecting specific types of manipulation [17], [18], [19]. Since manipulation type is unknown in the real world, it is important to develop generic models for multiple manipulation types.

Recently, some works that can detect multiple manipulation types have been proposed. ManTra-Net [20] formulated the forgery localization problem as a local anomaly detection problem and can detect multiple manipulation types. SPAN [21] utilized a pyramid of local self-attention blocks to model spatial relationships among image patches at multiple scales. ManTra-Net and SPAN have limitations in generalization because of failing to make full use of the spatial

correlation. Yang et al. [34] constructed a constrained region-CNN, which utilized an attention region proposal network for coarse localization and fused low-level and high-level information to obtain refined tampered regions. Zhuang et al. [30] proposed a dense fully convolutional network, where dense connections and dilated convolutions were adopted for achieving better localization performance. Zeng et al. [35] proposed a proxy proposal contrastive learning task to exploit relationships of local features and obtain more generalizable features. Most of the above methods utilize the idea of segmentation to deal with the problem of image manipulation detection. However, Zhang et al. [36] first implemented image manipulation detection from a regression perspective and proposed a regression network based on CatmullRom splines. In addition, Ji et al. [37] and Xu et al. [38] utilize uncertainty learning to improve the generalization of image manipulation detection. Ji et al. [37] proposed an uncertainty estimation network to measure the uncertainty of data and model and improve the results of manipulation localization. Xu et al. [38] designed an uncertainty-supervised parallel network to avoid semantic noise while preserving more manipulation details. To fully employ the noise inconsistency between forged and real regions, Zhu et al. [39] proposed a two-step discriminative noise-guided method to enhance the noise inconsistency and improve the accuracy and robustness of forgery detection. Huang et al. [40] employed an RGB stream to extract both high-level and low-level tampering traces for coarse localization and adopted a noise stream to expose local noise inconsistency for fine localization. In this work, the proposed DAE-Net adopts a two-branch coding structure to extract semantic features and noise features at multiple scales. Different from existing methods that directly fuse features from different domains by summation or concatenation, which makes it difficult to achieve effective feature fusion, the proposed DAM module is employed to fuse semantic features and noise features by utilizing the self-attention mechanism in the spatial and channel dimensions, respectively. It can highlight the differences between tampered and real regions and obtain more discriminative feature representations.

B. Attention Mechanism

Attention mechanism [18], [22] has been applied to IMDL to further improve the feature representation with relatively low cost, among which spatial attention (SA) [41], channel attention (CA) [42], and self-attention [31] are more widely implemented. Liu et al. [41] proposed an attention-aware hierarchical-feature fusion module, which adopted position attention to fuse hierarchical features from two different domains. Ganapathi et al. [42] utilized CA to extract attention-aware multiresolution features in the spatial and frequency domains to increase generalization on unseen manipulations. Li et al. [33] proposed a spatial-channel fusion excitation module to enhance the model's differentiation capability. A mutual attentive module was proposed to distill complementary signals from the foreground and the background features in [43]. Xu et al. [18] designed semantic-agnostic manipulation attention for mitigating the

effect of rich image semantics and devised the subtractive operation to promote the network to learn general features. Zhuo et al. [31] utilized a self-attention mechanism to obtain contextual dependencies of intrinsic inconsistency in tampered regions. Wang et al. [44] employed the self-attention mechanism in the transformer architecture to produce refined global representations for IMDL. To realize the fusion of features from different domains, Xia et al. [32] adopted a hierarchical feature fusion strategy, which utilized the convolutional block attention module (CBAM) and self-attention to fuse RGB and noise features hierarchically. Inspired by the above attention mechanism, we designed a DAM module to fuse features from different domains. Meanwhile, different from the decoding manner of existing methods, an attention fusion module (AFM) is designed in the mask generation process to fuse rich semantic information of different scales, enhancing the feature representation and improving the accuracy of manipulation localization.

C. Edge Supervision

To solve the problem of coarse and incomplete boundaries in current IMDL methods, recent works [3], [24] have introduced an edge supervision strategy to optimize the boundary of tampered regions. Lin et al. [28] enhanced fused tampering traces using the proposed edge artifact enhancement module and edge supervision strategy to find subtle edge artifacts hidden in images. Li et al. [3] applied morphological operations to extract multiscale edge information, which can reduce feature redundancy by producing higher resolution edges. Gao et al. [45] designed a multiscale edge-guided attention stream to obtain robust boundary information through a multiscale Sobel layer. Subsequently, the edge attention module accurately located the tampered regions based on the multiscale edge features. Li et al. [46] proposed the threshold-adaptive differentiable binarization edge algorithm to provide the learnable edges. Zeng et al. [47] proposed a dual-pathway multiscale network, which was mainly used for edge information extraction and adopted a dual-path structure to align visual features in RGB space and edge features in LAB space. MVSS-Net [24] introduced a Sobel layer to identify candidate edge pixels and reweighted the edge features by attention maps for accurate edge localization. Therefore, inspired by the edge supervision strategy of MVSS-Net, we introduce an ESM to optimize the boundary of tampered regions. In contrast to existing methodologies, we exploit the Sobel layer to identify the edge pixels at different scales. Then, the edge enhancement block is designed to enhance the edge regions and retain more detailed information.

III. PROPOSED METHOD

Our goals are to localize tampered regions and improve localization accuracy by modeling visual artifact features at multiple scales and modalities. In this section, the network structure of DAE-Net is described in detail. The DAE-Net consists of a two-branch structure: the RGB branch and the NB, which extract artifact features of different modalities. Moreover, we introduce a DAM module for feature fusion and

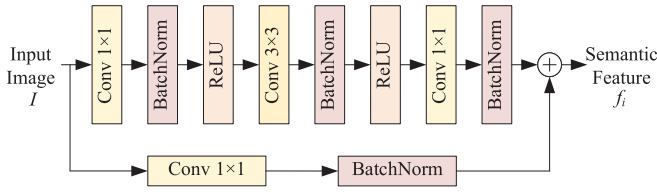


Fig. 3. Illustration of the basic structure of ResNet block.

a MGM for generating tampered masks. Finally, an ESM is designed to optimize the boundary of tampered regions. The details of the main network components are described below.

A. Overview

The overview of the proposed DAE-Net is shown in Fig. 2. An input image is denoted by $I \in \mathbb{R}^{H \times W \times 3}$, where H and W represent the height and width of the image, respectively. Our goal is to localize the tampered mask $M \in \mathbb{R}^{H \times W \times 1}$. As shown in Fig. 2, the whole network structure consists of five components: RGB branch, NB, DAM module, MGM, and ESM. We first employ the RGB branch and the NB to explore multimodal artifact features aiding the localization of tampered regions. The RGB branch is utilized to extract multiscale semantic features. The NB is employed to capture the fused global and local inconsistent noise features. Moreover, the DAM module is designed to fuse the semantic features and noise features to mine the complementary information between features and generate more generalized feature representations. Subsequently, the MGM is exploited to realize the localization of the tampered regions. During the mask generation process, the accuracy of the manipulation localization is improved by the FUM to make full use of semantic features at different scales. Finally, to optimize the boundary of tampered regions, the ESM is proposed to enhance the subtle edge features and guide the network to predict more refined tampered regions.

B. RGB Branch

The RGB branch aims to extract semantic features at multiple scales to provide better feature representations for image manipulation localization. The RGB branch is composed of four ResNet blocks in series, where the basic structure of each ResNet block is shown in Fig. 3. Each ResNet block contains a varying number of basic structures. Referring to the architecture of ResNet 50 [48], we set the number of basic structures in each block to be: 3, 4, 6, and 4, respectively. The CNN-based backbone is used to mine the local artifact features in the RGB domain of the input image. The input of the RGB branch is the forged image, denoted by $I \in \mathbb{R}^{H \times W \times 3}$. First, we perform the preprocessing operation to transform the input image via a 7×7 convolutional layer with a stride of 2 and 3×3 maximum pooling layer into the feature map $I' \in \mathbb{R}^{(H/4) \times (W/4) \times 64}$. To obtain multiscale features containing rich semantic content and spatial information for assisting the final tampered mask prediction, the preprocessed image is passed through four ResNet blocks in series to extract semantic features at four scales: $f_1 \in \mathbb{R}^{(H/4) \times (W/4) \times 256}$,

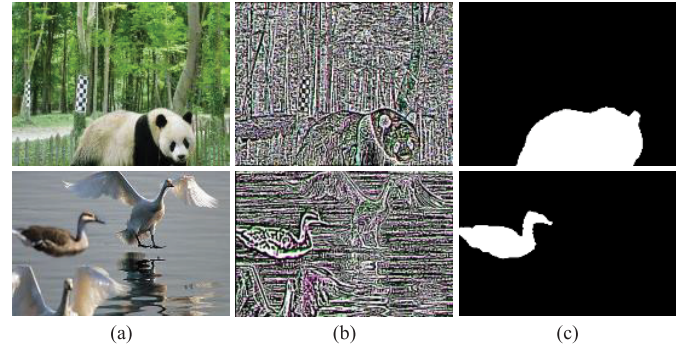


Fig. 4. Visualization results of extracted noise map by the constrained convolutional layer. (a) Original forged image. (b) Noise map. (c) Ground-truth binary mask.

$f_2 \in \mathbb{R}^{(H/8) \times (W/8) \times 512}$, $f_3 \in \mathbb{R}^{(H/16) \times (W/16) \times 1024}$, and $f_4 \in \mathbb{R}^{(H/16) \times (W/16) \times 2048}$. The above process is expressed as follows:

$$\{f_i, i = 1, 2, 3, 4\} \leftarrow \text{ResNet}\{\text{Maxpool}[\text{Conv}_{7 \times 7}(I)]\} \quad (1)$$

where $\text{Conv}_{7 \times 7}$ denotes the convolutional operation with filter size 7×7 and Maxpool denotes the maximum pooling operation.

C. Noise Branch

The RGB branch may ignore the invisible visual artifacts in the RGB domain when performing tampered features extraction, thereby reducing the generalization of manipulation localization. To improve the generalization ability and capture the invisible artifact features in the RGB domain, we introduce the NB to capture the noise features of the tampered images. Inspired by the SWT [29] that can extract global and local features, the NB adopts the SWT architecture as the backbone to extract noise features that fuse global and local inconsistencies. The NB consists of a noise map extraction block, patch embedding (PE), and SWT block.

The noise map extraction block, which consists of a constrained convolutional layer [27], [34], adaptively converts the image into noise maps. The constrained convolutional layer is implemented as a set of trainable filter kernels with constraints that enable the filter kernels to resemble high-pass filters. The extracted noise maps are shown in Fig. 4. The constraints on the filter kernels are expressed as follows:

$$\begin{cases} w_k(c, c) = -1 \\ \sum_{m, n \neq c} w_k(m, n) = 1 \end{cases} \quad (2)$$

where w_k is the weight of k th filter kernel, (c, c) is the center position coordinate of the filter kernel, and (m, n) is the noncenter position coordinate of the filter kernel.

The noise maps are fed into the network with a SWT as the backbone to extract the noise features that fuse the global and local inconsistencies. First, the patch partition module is utilized to divide the noise maps into nonoverlapping patches of size $N_p \times N_p$. In this study, N_p is set to 4. Then, a linear embedding layer is exploited to map the dimension of noise features to C , where the size of the noise features is $(H/4) \times (W/4) \times C$. Finally, four SWT blocks in series are used to

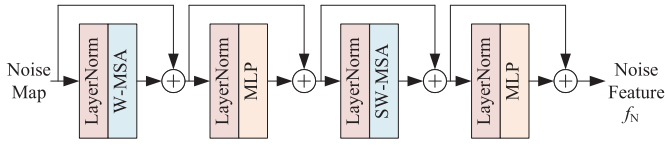


Fig. 5. Illustration of the basic structure of the SWT block.

extract the fused global and local inconsistent noise features from patches. The basic structure of the SWT block is shown in Fig. 5, which is made up of layer norm, window-based multi-head self-attention (W-MSA), multilayer perceptron (MLP), and shifted window-based MSA (SW-MSA). Each SWT block consists of a varying number of SWT basic structures and patch merging. The SWT block employs the shifted window self-attention mechanism to extract global and local noise inconsistencies and reduce the computational complexity. The patch merging performs a downsampling operation on the features to reduce the feature resolution. The self-attention matrix operation is formulated as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \quad (3)$$

where Q , K , and V are the query, key, and value obtained after the linear projection of the patches, respectively; B is the relative position bias; and d is the dimension of Q and K .

The noise patches are fed into four successive SWT blocks to obtain features $f_N \in \mathbb{R}^{(H/32) \times (W/32) \times 8C}$, which fuse global and local inconsistent information. The above process is expressed as follows:

$$f_N \leftarrow \text{SWT}\{\text{PE}[\text{NEB}(I)]\} \quad (4)$$

where NEB, PE, and SWT denote the noise extraction block, patch embedding, and successive swin transformer blocks, respectively.

D. DAM Module

To fully mine the complementary relationship between semantic features and noise features, we introduce a DAM module [49], which leverages the self-attention mechanism in the spatial and channel dimensions to achieve an effective fusion of different domain features. It solves the problem that the previous method [50] simply concatenates only in the channel dimension and cannot discover the complementary relationship of different domains. For the two modality features, semantic feature f_4 and noise feature f_N , the noise feature is upsampled twice and concatenated with the semantic feature in the channel dimension. The input features $f_1 \in \mathbb{R}^{(H/16) \times (W/16) \times 512}$ of the DAM module are obtained after a 1×1 convolutional layer.

The DAM module consists of two parallel branches: the SA branch and the CA branch. The DAM module structure is shown in Fig. 6. In the SA branch, long-range spatial context information is captured by applying the self-attention mechanism in the spatial dimension. The input features are passed through three 1×1 convolutional layers to obtain query features $f_Q \in \mathbb{R}^{(H/16) \times (W/16) \times 64}$, key features $f_K \in \mathbb{R}^{(H/16) \times (W/16) \times 64}$, and value features $f_V \in \mathbb{R}^{(H/16) \times (W/16) \times 512}$,

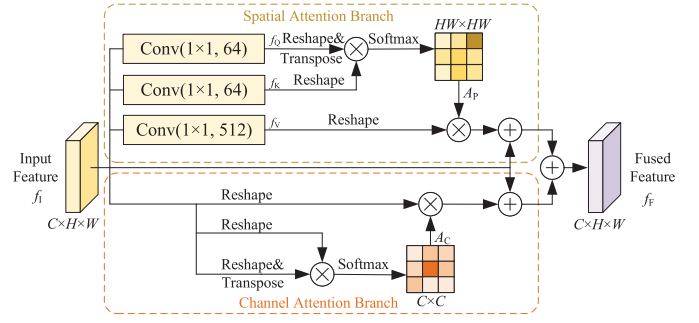


Fig. 6. Details of the DAM module, which contains the SA branch and CA branch.

where the query features and key features are downsampled in channel dimension to reduce the model computation. f_Q and f_V are reshaped to $f_Q \in \mathbb{R}^{N \times 64}$ and $f_V \in \mathbb{R}^{N \times 512}$, where $N = (H/16) \times (W/16)$. f_K is reshaped and transposed to obtain $f_K \in \mathbb{R}^{64 \times N}$. In the SA branch, SA matrix A_P of the feature maps f_1 is computed as follows:

$$A_P^{ij} = \frac{\exp(f_Q^i \times f_K^j)}{\sum_{i=1}^N \exp(f_Q^i \times f_K^j)} \quad (5)$$

where A_P^{ij} is the value of the i th row and j th column of the SA matrix A_P and N is the total number of pixels. SA feature maps f_P are computed as follows:

$$f_P = f_V \times A_P + f_1 \quad (6)$$

where f_1 denotes the input feature maps and f_V denotes the feature maps after 1×1 convolutional layer.

Similarly, in the CA branch, we reshape the input features to $f_R \in \mathbb{R}^{N \times 512}$ and transpose f_R to $f_T \in \mathbb{R}^{512 \times N}$. CA matrix A_C of the feature maps f_1 is computed as follows:

$$A_C^{ij} = \frac{\exp(f_T^i \times f_R^j)}{\sum_{i=1}^C \exp(f_T^i \times f_R^j)} \quad (7)$$

where A_C^{ij} is the value of the i th row and j th column of the CA matrix A_C and C is the number of channels. CA feature maps f_C are computed as follows:

$$f_C = f_R \times A_C + f_1. \quad (8)$$

Finally, the outputs of the two branches are aggregated to enrich the representation of artifact features and to highlight the differences between the tampered and authentic regions. The SA features f_P are summed with CA features f_C to obtain fusion features f_F , which is expressed as follows:

$$f_F = f_P + f_C. \quad (9)$$

E. Mask Generation Module

To solve the problem that previous methods [20], [24] cannot fully exploit the shallow features containing rich semantic information in the process of generating tampered mask, resulting in inaccurate prediction of tampered regions. As shown in Fig. 2, we adopt a progressive decoding manner to fully integrate low-level detailed features and high-level semantic features to improve the accuracy of manipulation

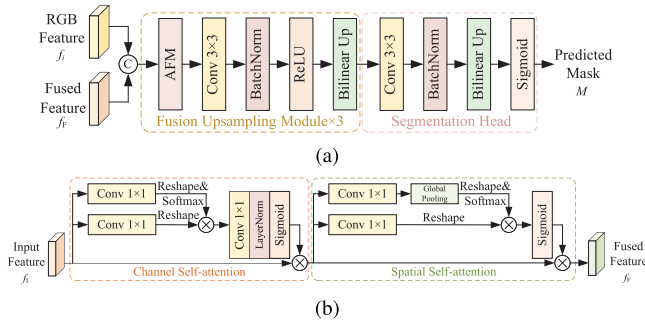


Fig. 7. Overall structure of MGM used in our proposed FUM for tampered regions detection. (a) MGM. (b) AFM.

localization. The MGM is composed of three FUMs and a segmentation head, as shown in Fig. 7(a). The FUM introduces an AFM to fuse multiscale semantic features, as shown in Fig. 7(b).

For the given fused feature f_F and multiscale semantic feature $\{f_i, i = 1, 2, 3\}$, they are first concatenated in the channel dimension to obtain feature f_S . Then, it is fed into the AFM to achieve effective feature fusion and enhance representation of the tampered features. Inspired by Liu et al. [51], we sequentially utilize self-attention in channel and spatial dimensions to fuse different features. The channel self-attention feature f_S^C is computed as follows:

$$\begin{cases} S_C = \text{Softmax}[\text{Conv}_{1 \times 1}(f_S)] \times \text{Conv}_{1 \times 1}(f_S) \\ f_S^C = \sigma\{\text{LN}[\text{Conv}_{1 \times 1}(S_C)]\} \otimes f_S \end{cases} \quad (10)$$

where $\text{Conv}_{1 \times 1}$ denotes the convolutional operation with filter size 1×1 , S_C is the channel self-attention, LN denotes the LayerNorm, σ is the Sigmoid function, and $\otimes$ denotes the elementwise multiplication.

The channel self-attention feature f_S^C is used as input to compute the spatial self-attention feature f_S^P , which is formulated as follows:

$$\begin{cases} S_P = \text{Softmax}\{\text{GP}[\text{Conv}_{1 \times 1}(f_S^C)]\} \times \text{Conv}_{1 \times 1}(f_S^C) \\ f_S^P = \sigma(S_P) \otimes f_S^C \end{cases} \quad (11)$$

where GP is the global pooling operation and S_P is spatial self-attention.

The fused features are then passed through a 3×3 convolutional block and upsampled to twice the original size by bilinear interpolation, thus completing a fusion upsampling operation. The above process is expressed as follows:

$$f_F^1 = \text{BU}\{\text{ReLU}[\text{BN}(\text{Conv}_{3 \times 3}(f_S^P))]\} \quad (12)$$

where $\text{Conv}_{3 \times 3}$, BN, ReLU, and BU denote the convolutional operation with filter size 3×3 , a batch normalization layer, an ReLU activation function, and bilinear interpolation upsample, respectively.

To fuse the three scales of semantic features $\{f_i, i = 1, 2, 3\}$, we adopt a progressive decoding manner and perform three fusion upsampling operations to obtain fused features $\{f_F^i, i = 1, 2, 3\}$ at different scales, as shown in Fig. 2. The obtained fused feature f_F^3 are input into the segmentation head

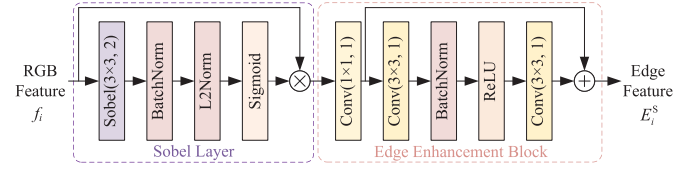


Fig. 8. Illustration of ESM, which is composed of the Sobel layer and edge enhancement block for tampered edge detection.

to predict the final tampered mask $M \in \mathbb{R}^{H \times W \times 1}$. This process can be represented as follows:

$$M = \sigma\{\text{BU}[\text{BN}(\text{Conv}_{3 \times 3}(f_F^3))]\} \quad (13)$$

where the bilinear interpolation upsampling multiple is set to 2.

F. Edge Supervision Module

To address the problem of rough and incomplete boundaries for tampered region localization, existing methods [3], [24], [28] extract edge features to optimize the boundary of tampered regions. Inspired by Dong et al. [24], the ESM is designed to assist in the localization of tampered regions. The ESM consists of a Sobel layer and an edge enhancement block, as shown in Fig. 8. The ESM takes the multiscale semantic features extracted by the RGB branch as inputs, which contains four different scales of inputs in total. Therefore, the ESM is designed as four parallel Sobel branches to extract the edge information of the four scales of semantic features. The edge strategy is implemented to optimize the boundary of tampered regions.

For the given multiscale semantic feature $\{f_i, i = 1, 2, 3, 4\}$, we first extract the edge features with the Sobel convolutional layer, which is initialized using the Sobel operator. The multiscale semantic features are fed into the Sobel convolution layer to identify the edge pixels. Then, the attention maps of the edge pixels are obtained after BN layer, L2Norm, and sigmoid function. After elementwise multiplication of the attention maps with the input feature maps, multiscale edge features are obtained. This process is expressed as follows:

$$E_i = \sigma\{L_2[\text{BN}(\text{Sobel}(f_i))]\} \otimes f_i \quad (14)$$

where $\{E_i, i = 1, 2, 3, 4\}$ are the edge features; L_2 and Sobel denote the L2Norm layer and Sobel convolutional layer, respectively.

Subsequently, the edge enhancement module is exploited to enhance the edge features E_i so that the extracted edge features retain more edge detail information. Specifically, a 1×1 convolutional layer is first used to change the number of channels. Then, the edge features are enhanced by two 3×3 convolutional layers. The enhanced edge features are represented as follows:

$$E_i^S = \text{Conv}_{3 \times 3}\{\text{ReLU}[\text{BN}(\text{Conv}_{3 \times 3}(E_i))]\} + E_i \quad (15)$$

where $\{E_i^S, i = 1, 2, 3, 4\}$ denote the enhanced edge features.

Finally, the four enhanced edge features are upsampled to the same size as the tampered image and are concatenated in

the channel dimension to fuse different scales of edge information. The edge mask is predicted with the edge prediction head. This process is represented as follows:

$$M_E = \text{Conv}_{1 \times 1} [\text{Cat}(E_i^S)_{i=1}^4] \quad (16)$$

where $\text{Cat}(\cdot)$ denotes the feature concatenation along the channel dimension. The network performs supervised training by calculating the loss between the edge prediction mask and the authentic edge mask to optimize the boundary of tampered regions.

G. Loss Function

During the end-to-end training of the network, we consider a multitask supervision strategy, i.e., tampered mask supervision and edge supervision. The objective of the tampered mask supervision is to enhance the model's sensitivity to boundary localization. Consequently, two loss functions were established to optimize the model's parameters: the tampered region loss L_M and the edge loss L_E .

1) *Tampered Region Loss*: The prediction of the tampered region can be viewed as a binary classification of pixels. Thus, the tampered region loss is generally computed using binary cross entropy L_B , which is denoted as follows:

$$L_B = -\frac{1}{m} \sum_{i=1}^m [y_i \log_2(\hat{y}_i) + (1 - y_i) \log_2(1 - \hat{y}_i)] \quad (17)$$

where m is the number of pixels and y_i and $\hat{y}_i$ are the ground-truth and prediction of the i th pixel, respectively. However, in the image tampering, the size of the tampered region is uncertain. Sometimes the tampered region is only a small part of the whole image, which suffers from the problem of sample unbalance. Inspired by Ding et al. [57], the dice loss L_D is introduced into the tampered region loss for optimization. The dice loss ignores a large number of background pixels in the calculation, so the problem of sample unbalance is well solved. The dice loss L_D can be expressed as follows:

$$L_D = 1 - \frac{2 \sum_{i=1}^m y_i \hat{y}_i}{\sum_{i=1}^m y_i^2 + \sum_{i=1}^m \hat{y}_i^2}. \quad (18)$$

The tampered region loss L_M is the loss between the ground-truth mask and the predicted mask and is expressed as follows:

$$L_M = \lambda L_B + \mu L_D \quad (19)$$

where L_B and L_D are the binary cross entropy loss and dice loss between the ground-truth mask and the predicted mask, respectively, and λ and μ are the two hyperparameters in the range of $[0, 1]$.

2) *Edge Loss*: The edge loss is the loss between the ground-truth boundary and the predicted boundary. Since the edge pixels of the tampered region are typically in the minority, we apply the dice loss which effectively copes with unbalanced positive and negative samples. The edge loss L_E is expressed as follows:

$$L_E = 1 - \frac{2 \sum_{i=1}^m e_i \hat{e}_i}{\sum_{i=1}^m e_i^2 + \sum_{i=1}^m \hat{e}_i^2} \quad (20)$$

where e_i and $\hat{e}_i$ refer to the value of the i th pixel in the ground-truth boundary and prediction boundary, respectively.

Overall, the total loss L_T for our model can be defined by combining the tampered region loss and the edge loss and is expressed as follows:

$$L_T = \gamma_M L_M + \gamma_E L_E \quad (21)$$

where γ_M and γ_E are the weights of the tampered region loss and edge loss, respectively.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we evaluate our model quantitatively and qualitatively. The experimental setup is introduced in Section IV-A, and the comparison results with the SoTA methods are presented in Section IV-B. Subsequently, we perform an ablation study to verify the effectiveness of different components and losses in Section IV-C. Furthermore, a robustness study is performed to evaluate the robustness of the model against different postprocessing attacks and specific conditions in Section IV-D. Finally, we show the visualization results in Section IV-E and compare the computational complexity of different methods in Section IV-F.

A. Experimental Setup

1) *Datasets*: The experimental images are mainly from five standard datasets, CASIA [52], NIST16 [53], Columbia [54], Coverage [55], and IMD2020 [56]. For a fair comparison, we adopt the most popular training-testing splitting configurations on these five datasets, as in [3], [22], and [44]. The details of these datasets are presented in Table II. We will briefly introduce these datasets as follows.

- 1) *CASIA [52]*: CASIA contains two versions, i.e., CASIA V1.0 and CASIA V2.0, where CASIA V1.0 has 920 tampered images and CASIA V2.0 has 5123 tampered images. CASIA provides spliced and copy-move images of various objects. We follow the methods in [3] and [44] and used CASIA V2.0 as the training set and CASIA V1.0 as the testing set.
- 2) *NIST16 [53]*: NIST16 is a challenging dataset, in which we perform extensive experiments. NIST16 includes 564 tampered images covering three manipulation types: copy-move, splicing, and removal. The ground-truth masks are provided.
- 3) *Columbia [54]*: Columbia contains only 180 spliced images. We follow the method in [3] and randomly use 130 images as the training set and the remaining 50 images as the testing set.
- 4) *Coverage [55]*: Coverage provides 100 images generated by copy-move techniques and also includes ground-truth masks.
- 5) *IMD2020 [56]*: IMD2020 includes 2010 real-life tampered images collated from the Internet and includes all three manipulation types.

2) *Compared Methods*: We compare DAE-Net with SoTA methods, which can be divided into two categories: unsupervised methods, e.g., CAF1 [9], NOI1 [10], and DNN-based methods, e.g., ManTra-Net [20], SPAN [22], DenseFCN [30],

TABLE II
TRAINING-TESTING SPLIT AND DESCRIPTION OF STANDARD DATASETS. ✓ AND ✗ INDICATE WHETHER OR NOT THE MANIPULATION TYPE IS INVOLVED

Dataset	Training-Testing Split		Involved Manipulation Type			Is Processed	Total
	Training	Testing	Copy-move	Splicing	Removal		
CASIA [52]	5123	921	✓	✓	✗	Yes	6044
NIST16 [53]	404	160	✓	✓	✓	Yes	564
Columbia [54]	130	50	✗	✓	✗	No	180
Coverage [55]	75	25	✓	✗	✗	Yes	100
IMD2020 [56]	1610	400	✓	✓	✓	Yes	2010

SATFL [31], MVSS-Net [24], DMFF-Net [32], and SF-Net [33]. These methods are briefly described as follows.

- 1) *CFA1* [9]: CFA1 performs image forgery localization by analyzing CFA artifacts.
- 2) *NOI1* [10]: NOI1 leverages noise inconsistency for manipulation detection.
- 3) *ManTra-Net* [20]: ManTra-Net detects and localizes tampered regions by a local anomaly detection network.
- 4) *SPAN* [22]: SPAN utilizes a pyramid structure and models the relationship among image blocks by local self-attention blocks.
- 5) *DenseFCN* [30]: DenseFCN leverages a fully convolutional encoder-decoder structure with dense connections and dilated convolution for achieving better localization performance.
- 6) *SATFL* [31]: SATFL employs a self-attention mechanism to localize tampered regions, where the self-attention module is based on a channel-wise high-pass filter block.
- 7) *MVSS-Net* [24]: MVSS-Net jointly utilizes multiview input, boundary artifacts, and the holistic information in an end-to-end manner to extract semantic-agnostic and generalizable features.
- 8) *DMFF-Net* [32]: DMFF-Net uses a graded feature fusion strategy and a multisupervision strategy to achieve accurate tampered region localization.
- 9) *SF-Net* [33]: SF-Net utilizes a feature enhancement module to enhance the fine-grained features in RGB stream and adopts a fusion excitation strategy to effectively fuse spatial and channel features.

3) *Implementation Details*: The proposed model is implemented based on Pytorch and is trained over the platform with two NVIDIA GeForce RTX 2080Ti GPUs. Due to the memory limitations of the computing devices, all images are resized to 512×512 . We use Adam for optimization with a learning rate of $1e-5$ and a weight decay of $5e-5$. We train the whole model for 60 epochs with a batch size of 8. In our tampered region loss function, the hyperparameters λ and μ are set to 0.5. In the total loss function, the tampered region loss weight γ_M and the edge loss weight γ_E are set to 0.8 and 0.2, respectively.

4) *Evaluation Metrics*: We evaluate the performance of the proposed method for image manipulation localization at the pixel level. Following previous works [2], [3], [24], pixel-level

F_1 score and area under the receiver operating curve (AUC) are adopted to evaluate the performance of binary classification for every pixel. Both F_1 score and AUC are in the range of [0, 1]. Larger values of them indicate better localization performance of the model. F_1 score is calculated as follows:

$$F_1 = \frac{2pr}{p+r} \quad (22)$$

where p and r represent the precision and recall, respectively. F_1 is the harmonic average of the precision and recall. F_1 and AUC are calculated on each image, respectively. The mean value of each metric for all images in a dataset is used to make comparisons in the following experiments.

B. Comparison With the SoTA Methods

We compare and analyze DAE-Net with the SoTA image manipulation localization methods on five benchmark datasets. The compared IMDL methods are divided into two categories: unsupervised methods, e.g., CAF1 [9], NOI1 [10], and DNN-based methods, e.g., ManTra-Net [20], SPAN [22], DenseFCN [30], SATFL [31], MVSS-Net [24], DMFF-Net [32], and SF-Net [33]. For a fair and reproducible comparison, the networks are trained using the same training strategy for methods with accessible source code. Their performance is then evaluated on five benchmark datasets. If the source code is not available, the reported results are from the relevant references. In particular, the reported results of ManTra-Net [20], DenseFCN [30], MVSS-Net [24], DMFF-Net [32], and SF-Net [33] are obtained by retaining them with their public codes on our training samples. The reported results of SPAN [22] and SATFL [31] are obtained by considering the corresponding and related literature [3], [28]. We analyze the comparison results from both quantitative and qualitative perspectives. The generalization of the proposed model is discussed.

1) *Quantitative Comparison*: For a fair comparison between the proposed method and nine SoTA methods, we utilize F_1 and AUC to measure the difference between the ground-truth mask and the predicted mask. Table III and Fig. 9 show a quantitative comparison of the F_1 and AUC on five benchmark datasets, with the best performance bolded. It is seen that DNN-based methods show better performance compared with unsupervised methods. It is observed that the F_1 of the DAE-Net achieves optimal results on CASIA and Columbia, and suboptimal results on the remaining three

TABLE III

QUANTITATIVE COMPARISON OF DAE-NET WITH THE NINE SoTA METHODS ON FIVE PUBLIC DATASETS: CASIA, NIST16, COLUMBIA, COVERAGE, AND IMD2020. “-” INDICATES THAT THE RESULTS ARE NOT PROVIDED IN THE LITERATURE

Method	Category	CASIA		NIST16		Columbia		Coverage		IMD2020		Average	
		F_1	AUC	F_1	AUC	F_1	AUC	F_1	AUC	F_1	AUC	F_1	AUC
CFA1 [9]	Unsupervised	0.207	0.522	0.174	0.501	0.467	0.720	0.190	0.485	-	-	-	-
NOI1 [10]	Unsupervised	0.263	0.612	0.285	0.487	0.574	0.546	0.269	0.587	-	-	-	-
ManTra-Net [20]	DNN-based	0.223	0.648	0.462	0.795	-	0.824	0.283	0.777	0.265	0.785	-	0.799
SPAN [22]	DNN-based	0.213	0.709	0.582	0.961	0.815	0.936	0.325	0.791	-	-	-	-
DenseFCN [30]	DNN-based	0.209	0.631	0.704	0.954	0.710	0.881	0.185	0.754	0.286	0.723	0.419	0.789
SATFL [31]	DNN-based	0.246	0.697	0.613	0.937	0.804	0.892	0.347	0.767	0.300	0.796	0.462	0.818
MVSS-Net [24]	DNN-based	0.390	0.748	0.827	0.981	0.703	0.719	0.284	0.808	0.411	0.817	0.523	0.815
DMFF-Net [32]	DNN-based	0.386	0.791	0.843	0.976	0.837	0.945	0.282	0.727	0.328	0.784	0.535	0.845
SF-Net [33]	DNN-based	0.464	0.921	0.911	0.928	0.805	0.948	0.311	0.737	0.344	0.768	0.567	0.860
DAE-Net (ours)	DNN-based	0.494	0.805	0.871	0.984	0.872	0.973	0.330	0.741	0.338	0.826	0.581	0.866

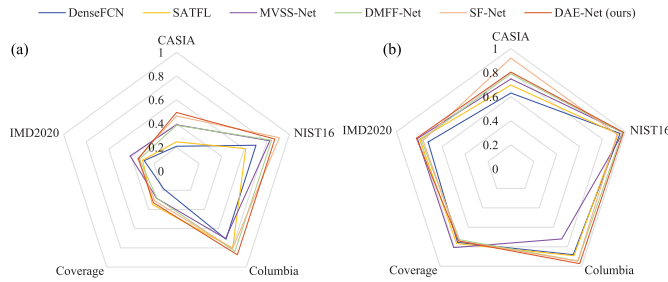


Fig. 9. Comparison results visualization with radar charts of the proposed method with five SoTA methods on five public datasets. (a) Results of F_1 . (b) Results of AUC.

datasets. The values of AUC achieve optimal results on NIST16, Columbia, and IMD2020. Moreover, the DAE-Net achieves the best results for the average of both F_1 and AUC on all five datasets, which are 0.581 and 0.866, respectively. On the CASIA dataset, F_1 of the proposed method is 0.271, 0.281, 0.285, 0.248, 0.104, 0.108, and 0.030 higher than that of ManTra-Net [20], SPAN [22], DenseFCN [30], SATFL [31], MVSS-Net [24], DMFF-Net [32], and SF-Net [33], respectively. This indicates that our method effectively improves the performance of manipulation localization by extracting noise and edge features, which ensures the generalization ability of the model. Our method obtains performance improvements of F_1 with gains of 4.4% on NIST16, 16.9% on Columbia, and 4.6% on Coverage when compared with MVSS-Net [24]. For the AUC measurement, DAE-Net achieves 82.6% on the real-world dataset IMD2020 and outperforms MVSS-Net and DMFF-Net by 0.9% and 4.2%, respectively. This suggests that our method adopts a progressive decoding manner to make full use of semantic features containing rich semantic information, which effectively improves the capability of manipulation localization and can generalize well to real-world datasets.

2) *Qualitative Comparison*: In this section, we present the qualitative results by comparing DAE-Net and the five most competitive methods: ManTra-Net [20], DenseFCN [30], MVSS-Net [24], DMFF-Net [32], and SF-Net [33]. Since the NIST16 dataset contains three manipulation types, we selected the NIST16 dataset for manipulation localization visualization.

Fig. 10 shows the qualitative visualization of manipulation localization for DAE-Net with five methods on the NIST16 dataset. It is observed that our method can localize tampered regions regardless of the manipulation types. The other five methods produce a large number of false alarms and incomplete predictions, i.e., tampered pixels cannot be localized accurately and completely. Among them, ManTra-Net [20] cannot achieve effective localization of removal manipulation type as it fails to extract the more generalized tampered features. Moreover, ManTra-Net [20] and DenseFCN [30] do not extract the edge features, resulting in inferior boundary localization compared with the proposed method. MVSS-Net [24] suffers from the problem of missing detection of tampered regions because it directly upsamples the extracted high-level tampered features during the mask generation process, making it difficult to effectively recovery the tampered regions. DMFF-Net [32] and SF-Net [33] still suffer from the loss of tampered regions and poor boundary localization. These localization visualization results can be further verified that the proposed method could localize more completely tampered regions. The utilization of fully multiscale features, which contain rich semantic information, in a progressive decoding manner, yields more accurate manipulation localization results.

3) *Generalization Ability Evaluation*: We further evaluate the generalization ability to unseen datasets. We train the model on the CASIA V2.0 dataset and perform the evaluation on the other unseen datasets, including CASIA V1.0, NIST16, and Columbia. The experimental results are shown in Table IV. It is observed that there is a degradation in the detection performance on the unseen dataset due to the larger gaps between different datasets. However, compared with the remaining three methods, our model achieves better performance on three datasets, which proves that our model has better generalization ability. This is attributed to the fact that DAE-Net captures and fuses different forgery clues to obtain a more generalized feature representation. Our DAE-Net outperforms Mantra-Net [20], DenseFCN [30], and MVSS-Net [24] by 0.245, 0.140, and 0.029 on the unseen Columbia dataset in terms of AUC, respectively, which demonstrates the superior performance of the proposed model. However, the model's

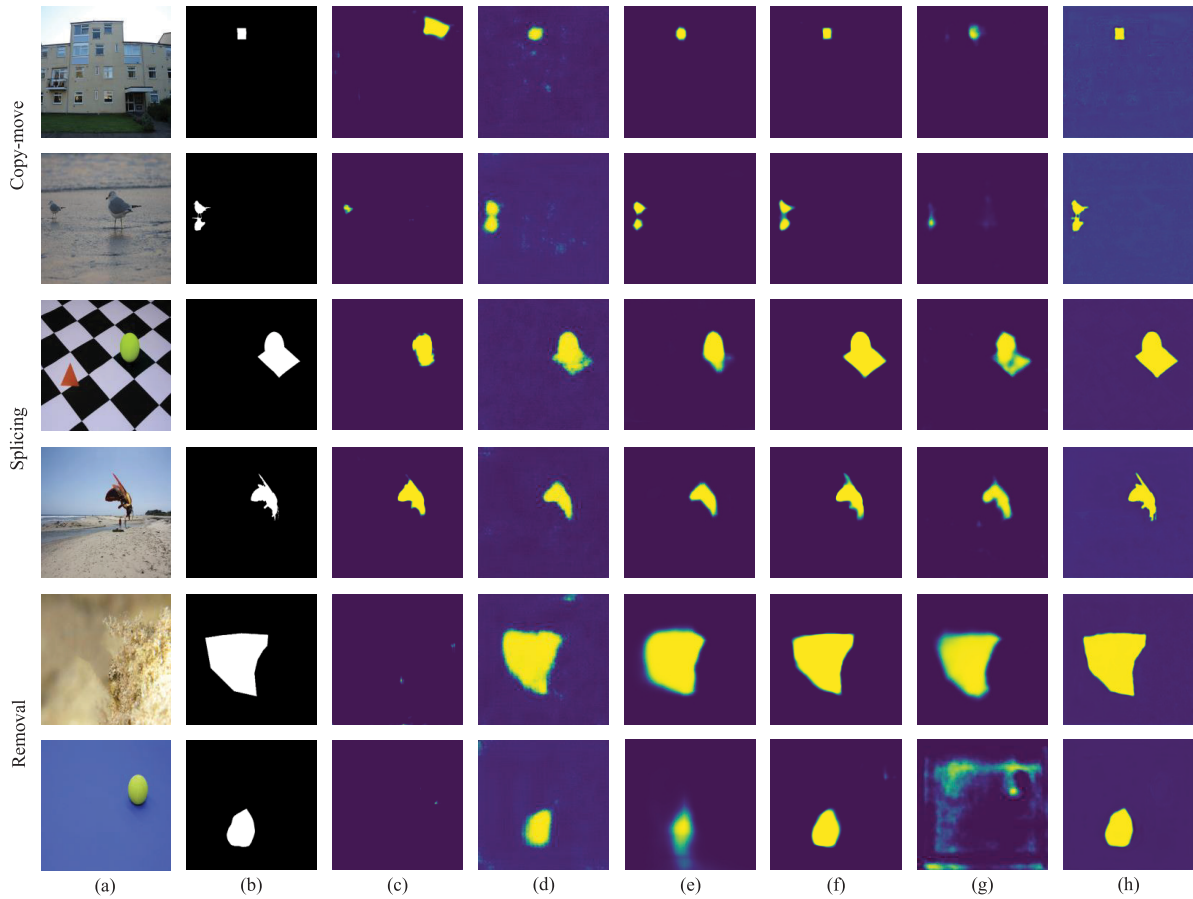


Fig. 10. Qualitative visualization results of the proposed DAE-Net compared with five SoTA methods against copy-move, splicing, and removal manipulations on NIST16 dataset. (a) Original forged image. (b) Ground-truth binary mask. (c) ManTra-Net. (d) DenseFCN. (e) MVSS-Net. (f) DMFF-Net. (g) SF-Net. (h) DAE-Net (ours).

TABLE IV

GENERALIZATION PERFORMANCE ON UNSEEN DATASETS. F_1 AND AUC ARE REPORTED IN THIS EXPERIMENT

Method	CASIA		NIST16		Columbia	
	F_1	AUC	F_1	AUC	F_1	AUC
ManTra-Net [20]	0.223	0.648	0.095	0.475	0.386	0.481
DenseFCN [30]	0.209	0.631	0.051	0.604	0.317	0.586
MVSS-Net [24]	0.390	0.748	0.246	0.643	0.471	0.697
DAE-Net (ours)	0.494	0.805	0.299	0.663	0.486	0.726

generalization for cross-dataset detection still need to be further improved.

C. Ablation Study

The proposed method designs the NB to capture fused global and local inconsistent noise features and an ESM to refine the boundary for localization of the tampered regions. Meanwhile, a progressive decoding manner is adopted to achieve the localization of the tampered region by FUMs. To evaluate the effectiveness of each module, we perform an ablation study on the NB, DAM, ESM, and MGM. Furthermore, we discuss the effect of different loss functions. All the ablation study in this section is conducted on the NIST16 or Columbia datasets.

1) Effectiveness of Different Components: First, we test the performance with/without NB, and the results are shown in Table V. It is observed that the performance of DAE-Net is significantly improved by adding semantic-agnostic noise features. Subsequently, we conducted an ablation study on the effectiveness of the MGM module, and the results are shown in Table VI. The localization accuracy of the model is improved after adding MGM. This is attributed to the fact that MGM utilizes the AFM to fuse multiscale semantic features in a progressive manner, effectively reducing the false alarm rate of tampered regions. In addition, Table VII demonstrates the effectiveness of DAM. We compare three attention fusion methods, such as SA, CA, and CBAM [58]. The DAM outperforms SA, CA, and CBAM in the NIST16 and Columbia datasets. This is because DAM utilizes the global information in spatial and channel dimensions to mine the complementary relationship between semantic features and noise features to obtain discriminative feature representation.

Finally, we construct five model variants, where the baseline model contains only the RGB branch. The remaining four model variants are then constructed by sequentially adding NB, DAM, ESM, and MGM to the baseline model. All model variants employ the same training settings. F_1 and AUC for five model variants on the NIST16 dataset are presented in Table VIII. It is observed that the proposed components are

TABLE V

ABLATION STUDY OF NB ON NIST16 AND COLUMBIA DATASETS. F_1 AND AUC ARE REPORTED IN THIS EXPERIMENT

Model Variant	NIST16		Columbia	
	F_1	AUC	F_1	AUC
w/o NB	0.621	0.947	0.792	0.938
w/ NB + w/o Noise Map	0.734	0.963	0.822	0.945
w/ NB + Noise Map	0.871	0.984	0.872	0.973

TABLE VI

ABLATION STUDY OF MGM ON NIST16 AND COLUMBIA DATASETS. F_1 AND AUC ARE REPORTED IN THIS EXPERIMENT

Model Variant	NIST16		Columbia	
	F_1	AUC	F_1	AUC
w/o MGM	0.601	0.969	0.738	0.915
w/ MGM + w/o AFM	0.773	0.979	0.808	0.949
w/ MGM + AFM	0.871	0.984	0.872	0.973

TABLE VII

ABLATION STUDY OF DIFFERENT ATTENTION FUSION STRATEGIES ON NIST16 AND COLUMBIA DATASETS. F_1 AND AUC ARE REPORTED IN THIS EXPERIMENT

Model Variant	NIST16		Columbia	
	F_1	AUC	F_1	AUC
w/o DAM	0.666	0.944	0.773	0.927
w/ SA	0.784	0.955	0.832	0.956
w/ CA	0.741	0.953	0.816	0.942
w/ CBAM [58]	0.802	0.967	0.853	0.964
w/ DAM	0.871	0.984	0.872	0.973

effective and significantly improve the F_1 and AUC. Compared with the baseline, the NB is designed to extract the fused global and local inconsistent noise features in baseline + NB. F_1 of the baseline + NB is improved by 4.5%. Baseline + NB + DAM + ESM further improves both F_1 and AUC by introducing an ESM. It verifies that ESM can extract the subtle edge artifacts to optimize the boundary of the tampered region. Finally, we utilize a specially designed MGM to localize the tampered region by FUM in a progressive decoding fashion, obtaining the optimal performance with 0.871 in F_1 and 0.984 in AUC on NIST16 dataset. In addition, we visualize the localization results for different model variants. The corresponding results are shown in Fig. 11. The NB and DAM can basically complete the overall localization of the tampered region, which significantly improves the accuracy of manipulation localization. The ESM effectively assists the localization of the tampered boundary as well as the MGM adopts progressive decoding to solve the problem of insufficient tampered region segmentation.

2) *Effectiveness of Different Losses*: We discuss the effect of different loss functions and hyperparameters in (21). Six combinations of loss function hyperparameters, ($\lambda_M = 1, \lambda_E = 0$), ($\lambda_M = 0.8, \lambda_E = 0.2$), ($\lambda_M = 0.6, \lambda_E = 0.4$), ($\lambda_M = 0.5, \lambda_E = 0.5$), ($\lambda_M = 0.4, \lambda_E = 0.6$), and ($\lambda_M = 0.2, \lambda_E = 0.8$), are set to observe the experimental results, as shown in Table IX. It is observed that when ($\lambda_M = 0.8$ and $\lambda_E = 0.2$),

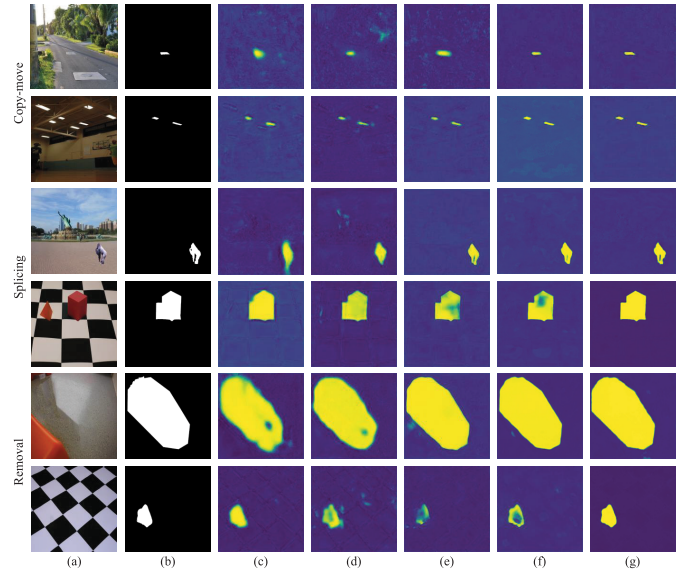


Fig. 11. Qualitative visualization results of different model variants against copy-move, splicing, and removal manipulations on NIST16 dataset. (a) Original forged image. (b) Ground-truth binary mask. (c) Baseline. (d) Baseline + NB. (e) Baseline + NB + DAM. (f) Baseline + NB + DAM + ESM. (g) DAE-Net.

the model achieves better performance. Therefore, our model choose ($\lambda_M = 0.8$ and $\lambda_E = 0.2$) as the default parameter setting for the experiment. Meanwhile, compared with only binary cross entropy loss L_B , adding dice loss L_D to tampered region can further improve the performance with gains of 13% in F_1 and 2.9% in AUC. This result shows that dice loss can effectively solve the problem of sample imbalance. On this basis, an edge loss L_E is introduced to optimize the boundary of tampered regions. By integrating all loss components, F_1 and AUC achieve the best results.

D. Robustness Study

When the tampered images are delivered on the Internet, they may undergo some image postprocessing operations that inevitably weaken or hide the tampering artifacts, which in turn affects the detection performance of the model. In this section, we further investigate the impact of robustness under different image postprocessing operations and different sizes of tampered areas. In addition, the performance of the model in specific conditions is discussed.

We apply three image postprocessing operations on the NIST16 dataset: Gaussian blur with a kernel size k , JPEG compression with a quality factor q , and Gaussian noise with a variance σ to evaluate the robustness of DAE-Net. We compared the robustness of DAE-Net with ManTra-Net [20], SPAN [22], DenseFCN [30], and MVSS-Net [24] on these corrupted data. The comparison results of F_1 and AUC between the proposed method and the SoTA methods on corrupted NIST16 dataset are shown in Fig. 12, where the values of kernel size k , quality factor q , and variance σ are [3, 9, 15], [50, 75, 100], and [3, 9, 15], respectively. It is observed that whichever image post-processing operations are used, the F_1 and AUC of the proposed method are higher than those of the other four SoTA methods. It indicates that

TABLE VIII

QUANTITATIVE COMPARISON OF DIFFERENT MODEL VARIANTS ON NIST16 DATASET. ✓ INDICATES THAT THE MODULE IS INVOLVED

Model Variant	Component					NIST16	
	RGB	NB	DAM	ESM	MGM	F_1	AUC
Baseline	✓					0.621	0.947
Baseline+NB	✓	✓				0.666	0.944
Baseline+NB+DAM	✓	✓	✓			0.722	0.965
Baseline+NB+DAM+ESM	✓	✓	✓	✓		0.773	0.979
Baseline+NB+DAM+ESM+MGM	✓	✓	✓	✓	✓	0.871	0.984

TABLE IX

ABLATION STUDY WITH DIFFERENT LOSSES AND WEIGHTS OF LOSS FUNCTION ON NIST16 DATASET. F_1 AND AUC ARE REPORTED IN THIS EXPERIMENT

Hyperparameter	Loss Function		NIST16	
	L_M	L_E	F_1	AUC
$\lambda_M = 1, \lambda_E = 0$	L_B	-	0.622	0.887
	$L_B + L_D$	-	0.677	0.903
$\lambda_M = 0.8, \lambda_E = 0.2$	L_B	L_E	0.741	0.955
	$L_B + L_D$	L_E	0.871	0.984
$\lambda_M = 0.6, \lambda_E = 0.4$	L_B	L_E	0.657	0.947
	$L_B + L_D$	L_E	0.762	0.959
$\lambda_M = 0.5, \lambda_E = 0.5$	L_B	L_E	0.652	0.954
	$L_B + L_D$	L_E	0.733	0.961
$\lambda_M = 0.4, \lambda_E = 0.6$	L_B	L_E	0.620	0.940
	$L_B + L_D$	L_E	0.703	0.956
$\lambda_M = 0.2, \lambda_E = 0.8$	L_B	L_E	0.624	0.923
	$L_B + L_D$	L_E	0.710	0.943

our method has better robustness against multiple distortion attacks. The performance of the SoTA methods is decreased when dealing with the postprocessed images, because of failing to effectively extract the tampering traces hidden by the postprocessing operations. However, the proposed method extracts complementary information on semantic features and noise features to enhance the tampering feature representation. Meanwhile, the edge supervision strategy is utilized to extract weakened edge artifacts to assist the localization of tampered regions. Since most of the postprocessing operations have a severe impact on the NB, we fuse multiscale semantic features in a progressive decoding manner to reduce the false and missed alarm rates for the localization of tampered regions. In summary, the strategy of fusing different tampered features (i.e., semantic features, noise features, and edge features) and progressive mask generation can effectively handle postprocessing attacks.

To test the model's performance in detecting different sizes of tampered areas, two types of tampering datasets are constructed, i.e., the large tampered area dataset and the small tampered area dataset. These two datasets each contain 100 images from the five datasets mentioned above. The datasets in which the pixels of tampered areas account for less than 12% of the pixels of the whole image are defined as small tampered areas, which is mainly based on the definition of the relative size of small targets. The test results on the two datasets are shown in Table X. It is observed that the proposed method achieves the optimal detection results compared with

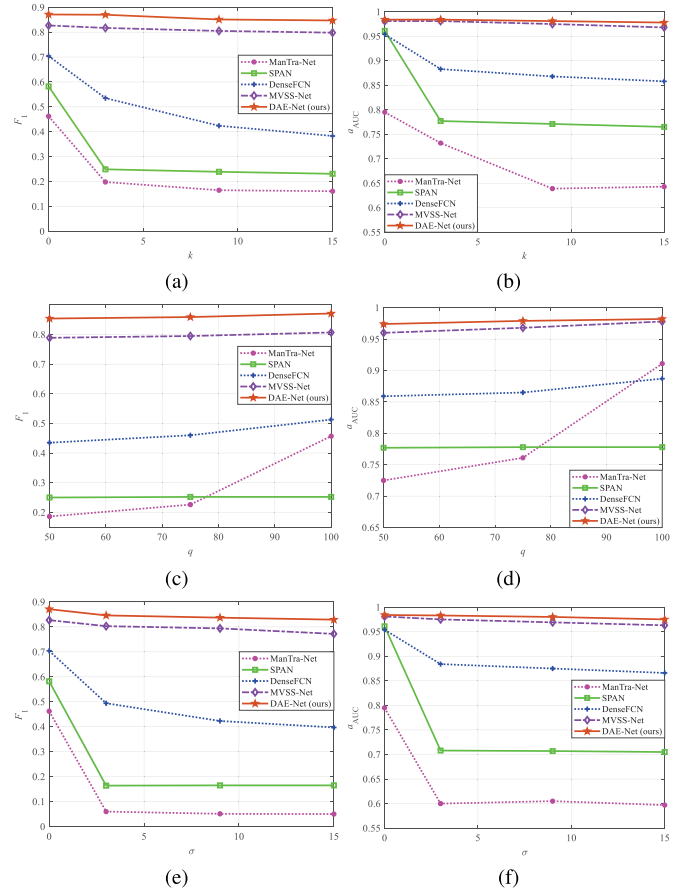


Fig. 12. Robustness comparison of the proposed DAE-Net with four SoTA methods under three image processing techniques. (a) and (b) Gaussian blur with kernel size k . (c) and (d) JPEG compression with quality factor q . (e) and (f) Gaussian noise with variance σ . The results of F_1 and AUC are reported, respectively.

four SoTA methods in terms of F_1 and AUC, which are 0.902 and 0.956 for the large tampered area dataset and 0.708 and 0.953 for the small tampered area dataset, respectively. In addition, we visualize the results of manipulation localization on both datasets, as shown in Fig. 13. In summary, the proposed method can effectively cope with different sizes of tampered areas, which is mainly attributed to the fact that the proposed model fully utilizes semantic features at different scales as well as edge information.

Moreover, the performance of the model under specific conditions (low resolution and high noise environment) is discussed. The experiments are performed on the NIST16 and

TABLE X

DETECTION PERFORMANCE OF DAE-NET IS COMPARED WITH FOUR SOTA METHODS ON LARGE TAMPERED AREA DATASET AND SMALL TAMPERED AREA DATASET. THE RESULTS OF F_1 AND AUC ARE REPORTED

Method	Large Tampered Area		Small Tampered Area	
	F_1	AUC	F_1	AUC
ManTra-Net [20]	0.572	0.725	0.108	0.447
DenseFCN [30]	0.506	0.709	0.245	0.711
MVSS-Net [24]	0.807	0.947	0.376	0.821
DMFF-Net [32]	0.873	0.943	0.561	0.871
DAE-Net (ours)	0.902	0.956	0.708	0.953

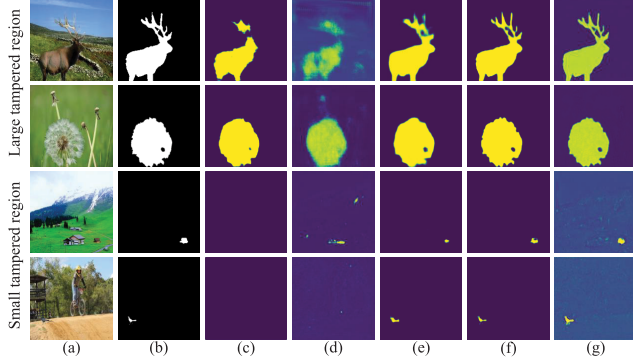


Fig. 13. Qualitative visualization results of the proposed DAE-Net on large tampered area dataset and small tampered area dataset. (a) Original forged image. (b) Ground-truth binary mask. (c) ManTra-Net. (d) DenseFCN. (e) MVSS-Net. (f) DMFF-Net. (g) DAE-Net (ours).

Columbia datasets. For the low resolution setting, the original image input 512×512 are reduced to 0.75, 0.5, and 0.25 times of the original, i.e., the input image sizes become 384×384 , 256×256 , and 128×128 , respectively. For the high-noise environment setting, different intensities of Gaussian noise are added to the input images, where the noise variance is set to 3, 9, and 15, respectively. The detection results of the proposed DAE-Net on NIST16 and Columbia in low resolution and high noise environments are shown in Table XI. It is observed that the proposed model is robust to high noise environments. For the low-resolution case, the performance of the model shows a decrease, indicating that the proposed method has some limitations for low resolution detection, which is mainly due to the fact that as the image resolution decreases, the forged boundaries and forged traces of the tampered image are blurred, making it difficult for the model to identify the tampered and authentic region efficiently.

E. Qualitative Visualizations

1) Qualitative Results on CASIA, Coverage, and Columbia:

To verify the generalization of the proposed DAE-Net on different datasets, the predicted masks on CASIA [52], Coverage [55], and Columbia [54] are shown in Fig. 14. It is observed that our method achieves accurate localization results for forged images on different datasets. No matter whether they are small manipulated targets or similar targets, our method can localize them accurately. It also illustrates that the proposed DAE-Net has superior generalization ability. The better localization results of DAE-Net are attributed to extensive

TABLE XI

DETECTION PERFORMANCE OF THE PROPOSED DAE-NET ON NIST16 AND COLUMBIA IN LOW RESOLUTION AND HIGH NOISE ENVIRONMENTS. THE RESULTS OF F_1 AND AUC ARE REPORTED

Specific Condition	Parameter	NIST16		Columbia	
		F_1	AUC	F_1	AUC
Original	-	0.871	0.984	0.872	0.973
Low Resolution	384×384	0.772	0.973	0.780	0.932
	256×256	0.754	0.965	0.660	0.855
	128×128	0.677	0.952	0.602	0.827
High Noise Environments	$\sigma = 3$	0.846	0.983	0.869	0.968
	$\sigma = 9$	0.837	0.980	0.834	0.960
	$\sigma = 15$	0.829	0.975	0.828	0.952

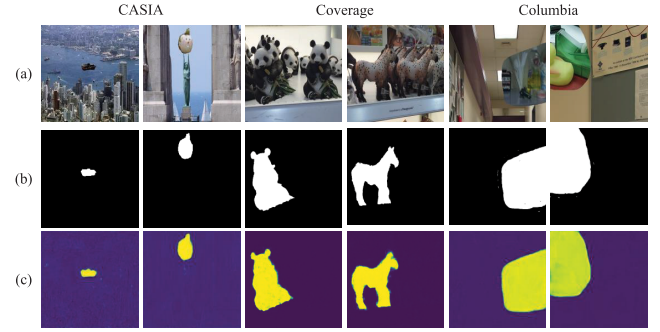


Fig. 14. Qualitative visualization results of the proposed DAE-Net on CASIA, Coverage, and Columbia. (a) Original forged image. (b) Ground-truth binary mask. (c) DAE-Net (ours).

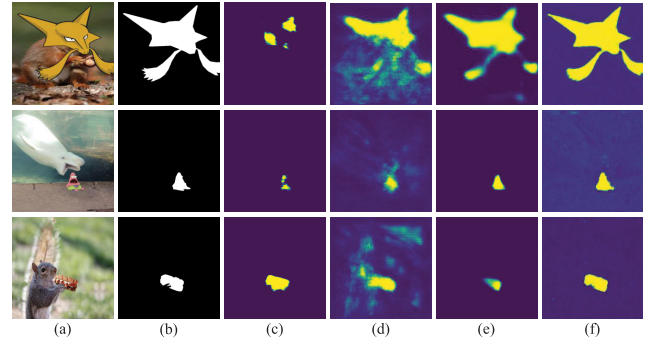


Fig. 15. Qualitative visualization results of the proposed DAE-Net on IMD2020 real-life dataset. (a) Original forged image. (b) Ground-truth binary mask. (c) ManTra-Net. (d) DenseFCN. (e) MVSS-Net. (f) DAE-Net (ours).

exploration of tampered features (i.e., semantic features, noise features, and subtle edge features) and the enhancement of subtle artifacts.

2) Qualitative Results on the IMD2020 Real-Life Dataset:

We compare the proposed DAE-Net with ManTra-Net [20], DenseFCN [30], and MVSS-Net [24] on the IMD2020 [56] real-life dataset. The qualitative visualization results are shown in Fig. 15. Compared with the other datasets, the IMD2020 [56] dataset contains more real forged scenarios, which can make the evaluation more convincing. As can be seen from Fig. 15, the proposed DAE-Net can segment the tampered regions completely and accurately. Nevertheless, ManTra-Net [20] cannot locate the tampered regions accurately and has poor generalization to the real-life dataset. Both DenseFCN [30] and MVSS-Net [24] can roughly localize the main body of the tampered regions, but the integrity of the

TABLE XII

COMPUTATIONAL COMPLEXITY COMPARED WITH THREE SoTA METHODS, WHERE “M” REPRESENTS MILLION, “G” REPRESENTS BILLION, AND “ms” REPRESENTS MILLISECONDS

Method	Params (M)	FLOPs (G)	Inference (ms)	Performance
ManTra-Net [20]	3.80	249.54	145.61	0.799
DenseFCN [30]	0.59	7.71	97.78	0.789
MVSS-Net [24]	142.78	163.38	108.57	0.815
DAE-Net (ours)	85.32	86.51	105.56	0.866

tampered regions and the accuracy of boundary localization are unsatisfactory. In addition, it is seen from the second row from Fig. 15, DAE-Net also identifies small tampered regions. These localization results indicate that the ESM and the progressive decoding manner can effectively capture edge artifacts and locate the tampered regions completely.

F. Computational Complexity Analysis

In this section, we compute and compare the complexity of different IMDL methods, i.e., ManTra-Net [20], DenseFCN [30], and MVSS-Net [24]. The computational complexity was calculated and compared with the above models in NVIDIA GeForce RTX 2080Ti GPU device. We compared model parameters, floating point operations (FLOPs), and inference time per image, and performance. Table XII presents the experimental results, where the inference time is the average of 150 randomly selected samples, and performance is the average AUC on the five datasets. From Table XII, it is observed that the proposed method consumes less inference time compared with ManTra-Net [20], with only 105.56-ms per image. Moreover, our model parameters are smaller compared with MVSS-Net. In summary, the proposed method achieves relatively small computational complexity, which means that our method can obtain a better experience in real scenarios compared with the SoTA methods.

V. CONCLUSION

In this work, we propose a novel DAE-Net to tackle the challenge of advanced image manipulation techniques. To obtain the invisible artifact information in the RGB domain, we extract the fused global and local inconsistent noise features in the noise domain as complementary information. Subsequently, we design a DAM module to enhance tampering feature representations by mining the complementary relationships of features from different domains. Furthermore, the progressive MGM is utilized to achieve manipulation localization at the pixel level, in which the designed FUM makes full use of the multiscale semantic features to improve the accuracy of manipulation localization. Finally, the ESM is designed to extract subtle edge features and refine the boundary of tampered regions. Extensive experiments on different datasets demonstrate the effectiveness and viability of the proposed method, which has better or comparable performance compared with the SoTA methods.

DAE-Net improves the accuracy of manipulation localization by extracting multiple tampered traces. However, the

generalization of the proposed method in cross-dataset detection still needs to be further improved. In further work, we will reconstruct the network architecture for strong generalization. On the one hand, the backbone of the NB is improved to extract global and local features by using a more lightweight architecture. On the other hand, more components can be designed to explore more tampered artifacts (e.g., compression artifacts and frequency domain features) for IMDL. Moreover, as AI-generated models evolve and a large number of AI images are produced, the compatibility of the model to detect AI-generated images can be further considered.

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